ITUH USED 27-10-2020PM PAPER

Code No: 117AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, October/November - 2020
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. What is meant by heuristic search technique? Give solution for blocks – world problem using heuristic search technique. [15]

2. Trace the Constraint satisfaction procedure solving the following cryptarithmetic problem:

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  CROSS
+ ROADS
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  DANGER
```
[15]

3.a) Differentiate Propositional logic and Predicate logic.
b) In Logic Programming how the problem is solved using "Proof by Refutation". [7+8]

4.a) When the extended semantic nets are useful? Explain with the example.
b) Illustrate with the example that how the knowledge is represented by using frames?[7+8]

5.a) What are the components in building an Expert System? With neat sketch explain how expert system works?
b) What is an expert shell? How it helps in building expert system? [7+8]

6.a) Consider the following set of propositions
i) A heart patient has spots
ii) A heart patient has measles
iii) A heart patient has high fever
iv) A heart patient has an allergy
v) A heart patient was recently bitten by a tick.
Create a network that defines the casual connections among these nodes. Convert it to a Bayesian network by constructing the necessary conditional probability matrix.
b) How Dempster-Shafer theory is helpful in representation of uncertainty? [7+8]

7. What is machine learning? What are the phases in it? How data is analyzed and classified through Support vector machine? [15]

8. How the knowledge is represented through semantic web? Explain with illustrations. [15]

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